

Abstract

Every day without fail, all around the world, there are thousands of weather balloons that are launched into the atmosphere for meteorological observation and research applications. These weather balloons are all comprised of a latex balloon that is filled with helium, and a radiosonde unit that hangs below the balloon itself. This radiosonde module plays a very important role in the measurement of atmospheric conditions within which the balloon resides, as it contains the numerous electrical components that give it the ability to measure characteristics of the atmosphere and the position in which the radiosonde finds itself in, as well as transmit those readings back to a ground station.

Currently, there are only a handful of companies around the world that design, manufacture, and sell radiosonde modules. These companies essentially have a monopoly on the entire market. One of the hindrances to companies/individuals designing and manufacturing their own radiosonde modules is the complicated process of transmitting the recorded information back to a ground station. Since the companies that manufacture the radiosonde modules have closed designs on all of their components, information is not widely available as to the exact inner workings of their designs for their transmission systems, other than the broad modulation schemes. It is therefore of paramount importance that alternative transmission schemes to the schemes that are employed by the few companies that manufacture radiosondes are explored if alternative designs to the mass-manufactured models are desired.

This report outlines research into an alternative transmission system based on the [Long Range \(LoRa\)](#) communication system. While LoRa is still a proprietary communication system, it is widely available due to its popularity among hobbyist communities, as well as its expanding footprint within the [Internet of Things \(IOT\)](#) community of devices. Reasons for popularity within these communities include the long-range transmission capabilities of the communication protocol, as well as its low cost, which makes it an excellent choice for the disposable radiosonde modules that are used for atmospheric observations.

A transmission and receiver system was designed, simulated, and built, based around the LoRa transmission protocol. Tests were then performed on the system to determine the optimal LoRa parameters for the usage of the protocol in radiosonde modules. From testing and simulation data, the best bandwidth was determined to be 62.5kHz, and the best spreading factor was determined to be 11.